

CLOUD COMPUTING

ASSIGNMENT#03

Name: **Uzair Ahmad**

Enrollment ID: **913**

Semester: **6th**

Email: ozyr.uc@gmail.com

Submitted to: **M. Waqas Sir**

**Department Of Software Engineering
University Of Malakand**

EduStream Pakistan: Cloud Architecture Design

Task 1: Virtualization Strategy

Hypervisors should manage the virtual machines because they isolate critical workloads and make the system easier to control during scaling. For EduStream, we use **VMs** for the main backend infrastructure where strong isolation is needed, and **containers** for lightweight services that need fast scaling and quick deployment.

- **Video processing:** Use containers, because encoding and streaming services need fast startup and easy horizontal scaling.
- **Backend APIs:** Use containers, because they change often and need continuous deployment.
- **Authentication service:** Use a VM or a very protected container setup, because it is small but highly critical and must stay secure.

Task 2: VM vs Container Trade-off Analysis

VMs provide stronger isolation, but they use more memory, boot slower, and make scaling harder. Containers are lighter, start faster, and work better for rapid deployment.

Main bottlenecks in VM-based deployment:

- High resource overhead.
- Slow startup during exam traffic spikes.
- Poor scaling when many users join at once.

Recommended hybrid approach:

- Use **VMs** for security-sensitive core services.
- Use **containers** for APIs, quizzes, dashboards, and video-related services.

Task 3: Microservices Architecture Design

Break the platform into these services:

- User/Auth Service
- Video Streaming Service
- Course Management Service
- Quiz Service
- Student Dashboard Service
- Notification Service

These services should communicate through REST APIs or internal service calls. Microservices are better than a monolith here because each part can scale separately, update independently, and recover without breaking the whole system. Containers support this by making each service easy to deploy, move, and scale.

Task 4: Resource Provisioning Strategy

Use **dynamic provisioning** instead of static provisioning because traffic changes a lot, especially during exam seasons. On normal days, the system should keep only the required minimum resources running. During peak time, it should automatically add more containers or VM instances.

Risks:

- **Over-provisioning:** Wastes money and resources.
- **Under-provisioning:** Causes slow response, buffering, and failed logins.

Task 5: Load Balancing Design

Place load balancers at the entry point before APIs, video servers, and quiz services. For **API requests**, use **round robin** or **least connections**. For **video streaming**, use **weighted load balancing** because some servers will handle heavier traffic than others.

- **Video streaming:** Weighted strategy fits best.
- **API requests:** Least connections works well because request load changes quickly.

Task 6: Scheduling Strategy

- **Video encoding:** Use **SJF** for short encoding jobs when possible, or **FCFS** for batch processing.
- **API requests:** Use **Round Robin** because many short requests must be treated fairly.
- **Authentication:** Use **Priority scheduling** because login should always be handled first.

Task 7: Failure Scenario Analysis

If one server crashes, the load balancer should stop sending traffic to it and move requests to healthy servers. If another server becomes overloaded, dynamic provisioning should launch more instances to share the load. Without load balancing, users would face downtime and slow responses. With poor scheduling, login and quiz requests may get delayed during peak usage.

Task 8: Critical Evaluation

Two weaknesses in this design are higher management complexity and higher monitoring needs. Another issue is that microservices can increase communication overhead between services. Improvements include better autoscaling, caching, and centralized monitoring. The main trade-off is that better performance costs more and adds complexity, but it is necessary for large-scale exam traffic.